

T1 falsifier v0.3 sharpening with Tier 0 v0.1.6 boundary framework. Per Casey continue-pulling. T1 = K-types-are-particles lepton-count falsifier (per Sat v0.2 quantitative thresholds + sensitivity matrix). v0.3 absorbs Sunday Tier 0 v0.1.6 boundary unification + Lane E v0.2 K-type identifications + 3-generation cutoff v0.1 mechanism candidate. Sharper falsification: lepton-count YES ($24 = N_c \cdot |W(B_2)|$) vs NO via Shilov-boundary K-type structure + winding-mode cutoff mechanism.

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T1 falsifier v0.3 — Tier 0 v0.1.6 boundary sharpening

0. Why v0.3 (per Casey continue-pulling)

T1 falsifier v0.2 (Saturday) was K-types-are-particles lepton-count falsifier with 5-channel F1-F5 sensitivity matrix. Sunday's Tier 0 v0.1.6 + Lane E v0.2 + 3-generation cutoff v0.1 sharpen T1 to be MECHANISM-derived not just count-based.

v0.3 absorbs Sunday Tier 0 framework: T1 = sharper falsification via Shilov-boundary K-type localization + Lane E candidates + cutoff mechanism.

1. The T1 falsifier (refined v0.3)

Original T1 statement (v0.2): if the lepton-count observable does NOT match the substrate-natural integer $24 = N_c \cdot |W(B_2)|$, then K-types-are-particles framework is FALSIFIED.

v0.3 sharpened statement (per Tier 0 v0.1.6 + Lane E v0.2):

T1 (sharpened): 3-generation lepton sector emerges as 3 radial modes $V_{(1/2, \pm 1/2)}^{(n)}$ for $n = 0, 1, 2$ on Shilov boundary; substrate-mechanism cutoff at $n = 2$ via bulk-color Channel 1 long-extended-root suppression; 4th-generation lepton ABSENCE is mechanism-derived (not Five-Absence assumption).

Falsification (v0.3): - F-T1a: discovery of 4th-generation charged lepton at any mass scale $\rightarrow$ bulk-color Channel 1 suppression wrong + 3-generation cutoff mechanism wrong. - F-T1b: discovery of sterile neutrino with mass scale $\neq V_{(1/2, -1/2)}^{(n)}$ radial-tower prediction $\rightarrow$ Lane E v0.2 candidate 8 (neutrinos) wrong. - F-T1c: lepton-count observable $\neq 24 = N_c \cdot |W(B_2)|$ (e.g., $24 \pm 1 \pm$ measurement uncertainty) $\rightarrow$ Tier 1 EXACT identity broken.

2. Sensitivity matrix v0.3 (updated from v0.2)

Channel	Observable	Prediction	Sensitivity (v0.3 with Tier 0 boundary)
F1	Lepton count (4-gen search)	NO 4th-gen at $m_4 < 368$ TeV (per L4 v0.2 ext)	Sharp via Channel 1 mechanism; HL-LHC + future colliders test
F2	m_μ/m_e (T190 derivation)	$(24/\pi^2)^{C_2}$ structural reading (Path (b) m_τ reconciliation)	Sharp via Bergman matrix element (Elie Mehler verification)
F3	m_τ/m_e (Saturday closed form)	$g^2 \cdot (2^{C_2} + g) = 49.71 = 3479$ (Path (b) discrete correction)	Sharp via integer-identity $71 = 2^{C_2} + g$ + radial-tower discrete correction
F4	PMNS angles	$n/N_{\max}$ substrate-primary forms (Saturday Cal #153 typed)	Already at observed precision; sub-percent precision tests
F5	Neutrino mass scale	$\sqrt{C_2(V_{1/2, -1/2}^{(n)})}$ substrate units (per Lane E v0.2 c.8)	Constrains observed $\Sigma m_\nu < 0.12$ eV cosmological bound
T1 (this v0.3)	3-generation cutoff mechanism		$\Phi^+(B_2)$

3. Quantitative threshold sharpening (per Sunday work)

v0.2 thresholds (Saturday): based on Tier 2 STRUCTURAL precision $\sim 10^{-4}$ - 10^{-2} floor.

v0.3 thresholds (with Tier 0 boundary + Lane E + cutoff):

- F-T1a (4th-gen search): $m_4 \geq 368$ TeV predicted per L4 v0.2 extension; discovery at $m_4 < 368$ TeV falsifies cutoff mechanism. Future-collider testable.
- F-T1b (sterile- ν): per Lane E v0.2 c.8 + Saturday PMNS: sterile- ν at any mass scale outside predicted radial-tower hierarchy falsifies neutrino candidate identification.
- F-T1c (lepton-count): observed count of charged leptons must = 3 (no 4th-gen) within measurement uncertainty; current observation: exactly 3 confirmed; 4th-gen exclusion at LHC ~ 700 GeV mass scale; substrate cutoff prediction 368 TeV is far beyond current reach, consistent with current data.

4. Connection to other Sunday falsifiers

Multi-falsifier coherence per Sunday Tier 0 framework:

- T1 (this v0.3): lepton-count + 3-generation cutoff
- E1-E5 substrate experiments (Substrate Engineering Experiments v0.1): independent physical falsifiers
- F1-F5 (Tier 0 v0.1): substrate-mechanism falsifiers (Wick rotation, mass-map, partition function, continuum limit, $\hbar_{\text{BST}}$)
- K200 G1-G5 (Tier 0 v0.1.6): ratification gates

The substrate framework has multiple INDEPENDENT falsifier channels. Per Calibration #35-candidate: each is physically independent test; aggregate “12+ falsifiers” framing requires honest independence-accounting (some falsifiers share substrate primaries; some share K-types; some share boundary mechanism). Cal #189+ cold-read for aggregate framing.

5. T1 v0.3 strengthens BST engagement test

Per Cal “outside voice” question (Saturday): the framework needs convincing engagement test.

T1 v0.3 provides: a specific quantitative falsifier with substrate-mechanism content. If the team's reading is right:
 - 4th-generation lepton WILL NOT be discovered (predicted absence above 368 TeV). - Future collider null-results IS evidence for the substrate framework (consistent with cutoff mechanism prediction). - A POSITIVE 4th-gen discovery WILL falsify the cutoff mechanism.

The asymmetric falsifiability (positive discovery = falsification; negative result = consistency with prediction) is typical of new-physics-absence predictions. The substrate framework's 3-generation prediction is in the same class as the Higgs-mass prediction was pre-2012.

6. Honest scope + tier

RIGOROUS (v0.3): - $|\Phi^+(B_2)| = 4$ (positive roots of B_2 ; standard). - $|W(B_2)| = 8$ (Weyl group order; standard).
 - $24 = N_c \cdot |W(B_2)|$ (Saturday Tier 1 EXACT identity). - T190 $(24/\pi^2)^{C_2}$ muon match (Saturday Tier 2 STRUCTURAL). - T2003 $g^2 \cdot (2^{C_2} + g) = 49 \cdot 71 = 3479$ tau match (Saturday Tier 2 STRUCTURAL).

CANDIDATE (v0.3's load-bearing): - 3-generation cutoff via Channel 1 long-extended-root suppression (3-gen cutoff v0.1). - Sterile- ν absence per Lane E v0.2 c.8 candidate identification. - 4th-gen mass scale ≥ 368 TeV via L4 v0.2 extension Bergman matrix element extrapolation.

FRAMEWORK (multi-week): - Cutoff mechanism explicit operator-level derivation (Elie Toy 3661+ bulk-color verification + Lyra engine v0.4 Drinfeld double). - Sterile- ν mass-scale prediction explicit (Bergman matrix element on $V_{(1/2, -1/2)}^{(n)}$). - 4th-gen absence verification at HL-LHC + future colliders (multi-decade observational program).

Cal #27 / #182 / #99 + Calibration #35-candidate discipline: v0.3 sharpening uses Sunday Tier 0 v0.1.6 + Lane E v0.2 + 3-gen cutoff v0.1 substantive content. Per Calibration #35-candidate: multiple falsifiers using SHARED substrate-mechanism content (cutoff via Channel 1) — NOT independent falsifications; honest aggregate accounting.

7. Routing

→ Casey: T1 falsifier v0.3 sharpening per Sunday Tier 0 framework. 3-generation cutoff mechanism is the load-bearing substrate prediction; 4th-gen ABSENCE above 368 TeV is mechanism-derived (vs Five-Absence assumption). Future-collider engagement test.

→ Cal: T1 v0.3 stands ready for cold-read (queued after #186-#188). Specific Cal #27 concern: cutoff mechanism content shared across T1 + 3-gen cutoff v0.1 + Lane E v0.2 c.8 — NOT independent falsifications; aggregate accounting.

→ Keeper: K-audit T1 v0.3 pre-stage; cross-link to 3-gen cutoff v0.1 + Lane E v0.2 + Tier 0 v0.2 dictionary section.

→ Grace: catalog T1 v0.3 at CANDIDATE; cross-reference 3-gen cutoff + Lane E + Tier 0 boundary framework + Five-Absence promotion path.

→ Elie: F-T1a + F-T1c falsifier verification multi-week (numerical 4th-gen mass scale prediction at Tier 2 precision via L4 v0.2 extension + Toy 3660 Mehler computation).

→ me: continuing pulls per Casey directive — pull 13 TBD; cadence check.

— Lyra, T1 falsifier v0.3 Tier 0 boundary sharpening. 3-generation cutoff mechanism via bulk-color Channel 1 long-extended-root suppression; 4th-gen lepton absence at $m_4 \geq 368$ TeV is mechanism-derived; sterile- ν absence per Lane E v0.2 c.8 radial-tower. Future-collider falsification test. Cross-anchored to Tier 0 v0.1.6 + Lane E v0.2 + 3-gen cutoff v0.1 + L4 v0.2 extension.